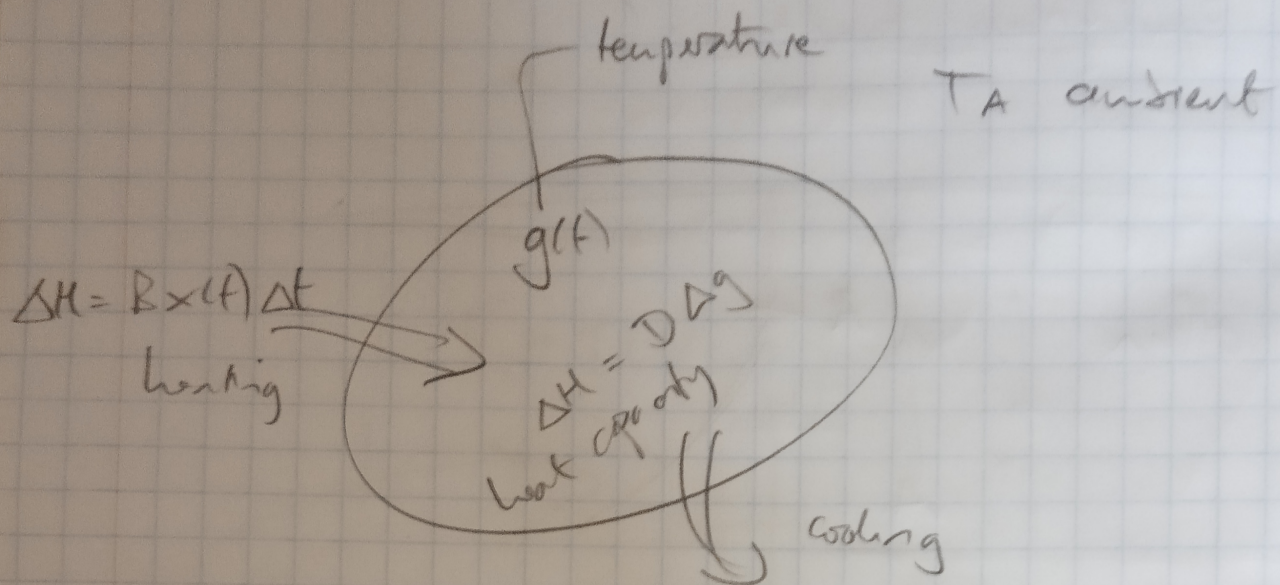




$$\Delta H = K(g(t) - T_a) \Delta t$$

$$B x(t) \Delta t - K(g(t) - T_a) \Delta t = D \Delta g$$

$$\text{let } y(t) = g(t) - T_a \Rightarrow y'(t) = g'(t)$$

Then

$$B x(t) - K y(t) = D y'(t)$$

power gain

heat transfer
coefficient

specific heat

More parameters than required for identification.
Normalise all by K , or just set $K=1$:

$$y(t) + D y'(t) = B x(t)$$

$$Y(s) + D(sY(s) - y(0^-)) = B X(s)$$

$$Y(s)(1 + Ds) = B X(s) + D y(0^-)$$

Zero input response: $x(s) = 0$

$$Y_{zi}(s) = \frac{Dy(0^-)}{1 + Ds} = \frac{y(0^-)}{s - (-1/D)}$$

$$y_{zi}(t) = y(0^-) e^{-1/D t}$$

Zero state response: $y(0^-) = 0$

$$Y_{zs}(s) = \frac{BX(s)}{1 + Ds} = \frac{B/D}{s - (-1/D)} X(s)$$
$$= H(s) X(s)$$

with $H(s) = \frac{B/D}{s - (-1/D)}$

Equivalently

$$y_{zs}(t) = L(t) * x(t)$$

with $L(t) = B/D e^{-1/D t}$

Full solution

$$y(t) = B/D e^{-1/D t} * x(t) + y(0^-) e^{-1/D t}$$

for $t \geq 0$

Special case $x(t) = 1$ (step)

$$y(t) = B(1 - e^{-1/D t}) + y(0^-) e^{-1/D t}$$

Special case $x(t) = 0$ (0/P)

$$y(t) = y(0^-) e^{-1/D t}$$